

Toward Behavioral Forensics of AIBO ERS-7 MIND 2: Recovery, Preservation, and Hypothesis-Driven Modeling of Persistent State

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Abstract

This paper reports a practical research program for studying higher-order behavior in Sony AIBO ERS-7 MIND 2 systems from preserved Memory Stick state. The work begins with forensic recovery of a non-booting Sony-formatted stick, establishes a reproducible restoration and preservation workflow, and then pivots into behavior forensics over persistent application state. We describe a host-side MIND 2 behavior simulator used as a hypothesis engine, identify candidate persistent-state levers including `STTLOG` row 0400, `STTLOG` row 1200, and specimen-side `IEG.CFG`, and show that these levers support distinct behavioral interpretations under a growing family of probe scenarios. The current evidence suggests that 0400 acts as a shutdown-side pivot, 1200 expresses socially-cued persistence, and specimen-side `IEG.CFG` expresses a more routine-preserving and multi-cycle persistence family. The contribution is not a claim of decoded Sony internal logic, but a reproducible methodology that links filesystem forensics, preserved media, host-side simulation, and live hardware testing.

1 Introduction

Behavior in legacy autonomous robots is often treated as opaque runtime magic, especially when vendor tooling and source-level documentation are incomplete or when deployed systems survive only as preserved media. For the Sony AIBO ERS-7 MIND 2 platform, this creates a two-part challenge. First, one must preserve and recover genuine bootable media accurately. Second, one must determine whether higher-order behavioral differences are represented in persistent state strongly enough to be isolated, transferred, and studied.

This paper presents the current state of an open research effort addressing both problems in sequence. The first milestone established that a Sony-formatted 32 MB class stick associated with an ERS-7 startup failure could be repaired by restoring a valid MIND 2 payload while preserving the underlying Sony media format. The second phase treated the repaired, working image as a behavioral specimen and developed a hypothesis-driven analysis loop over persistent files, transplanted state trees, and host-side simulated probe scenarios.

2 Research Goals

The work reported here is driven by three goals:

1. preserve accurate knowledge of genuine ERS-7 MIND 2 media and recovery;
2. identify which persistent files may encode behaviorally meaningful state;

3. build a reproducible path from preserved state to live-hardware interpretation.

The central research question is:

Where do higher-order MIND 2 behaviors live in persistent stick state, and to what extent are those behaviors reproducible?

3 Recovered Specimen and Preservation Milestone

The first major result was the repair and preservation of a Sony-formatted MIND 2 stick associated with a live ERS-7 startup failure. The forensic session established four points:

1. the captured medium had correct Sony 32 MB class low-level formatting;
2. the captured stick was non-bootable because it lacked `MEMSTICK.IND` and a complete `OPEN-R/` payload tree;
3. restoring the MIND 2 payload onto that Sony-formatted medium produced a stick that booted successfully on real hardware;
4. the repaired working stick was preserved as a new raw image for later study.

This matters methodologically because it separates media-format questions from payload-state questions. The subsequent behavior work is grounded in a specimen that is both preserved and known to boot.

4 Methodology

4.1 Behavior Forensics Workflow

The current workflow follows an observe–map–test loop:

1. observe a live behavior carefully and record it with a stable template;
2. map candidate behavioral differences to persistent files and file groups;
3. create one-variable test trees and compare their outcomes.

The preserved specimen image is held constant as the primary reference point. Candidate state changes are staged into sacrificial trees or test media rather than altering the preserved baseline.

4.2 Host-Side Simulator

Because direct robot testing is slow and expensive, a host-side MIND 2 behavior simulator was built as a hypothesis engine rather than a retail binary emulator. The simulator reads preserved persistent state, derives a symbolic behavior profile, and replays probe scenarios such as boot, interaction, disruption, shutdown, and sleep requests.

The simulator is explicitly limited:

- it does not execute Sony retail binaries;
- it is not evidence of exact Sony internal implementation;
- it is used to expose candidate structure and rank live-hardware tests.

4.3 WLAN Service Boundary

The current repo and recovered media also establish a useful boundary for network-facing work. On compatible ERS-7 WLAN configurations, the robot can expose live services including a wireless console in `WCONSOLE`, stock MIND 2 web content, and sample HTTP camera serving. In addition, decoded `MMFTP.CFG` content suggests bounded network-mediated content channels for specific payload families.

Methodologically, this is important because it distinguishes two different claims:

1. the running robot can expose observable and potentially interactive WLAN services;
2. the full Sony Memory Stick payload can be rewritten freely in place over WLAN.

The first claim is supported by current evidence. The second is not yet established. Accordingly, this work treats WLAN primarily as an observation, control, and service-interaction path rather than a proven general-purpose remote programming path for the retail stick.

4.4 Candidate Persistent-State Levers

Current analysis has focused on:

- `STTLOG` rows, especially 0400 and 1200;
- `IEG.CFG` as a structured behavioral state file;
- supporting state files including `PAT.LOG`, `FVAR`, `GVAR`, `AWAKING.CFG`, and `SIDRDATA.BIN`.

5 Current Findings

5.1 0400 as a Shutdown Pivot

The strongest current conclusion about `STTLOG` is that row 0400 acts as the dominant shutdown-side pivot in the host-side model. It does not explain all behavior by itself, but it is the clearest single lever for differentiating immediate compliance from resistance-oriented state.

5.2 1200 as a Social Persistence Signal

Row 1200 currently behaves less like a routine-preserving lever and more like a socially-cued bridge signal. Under probe scenarios involving high-salience interaction, it expresses short-lived but meaningful persistence. With later tuning, the model was able to sustain two cycles of social redirection after a strong cue while remaining weaker under lower-salience interaction.

5.3 Specimen-Side `IEG.CFG` as a Routine-Preserving Signal

`IEG.CFG` emerged as the strongest non-`STTLOG` behavioral candidate. The current specimen-side file differs sharply from the bundled baseline despite sharing the same size and `IEPR` header. When the body after offset `0x40` is treated as a structured grid, the baseline file has mixed non-zero values while the specimen-side file is dominated by repeated `48.0 + 0` cells and contains a long contiguous zero-second run.

This provided the structural basis for modeling `IEG.CFG` not as a simple boolean presence, but as a routine-rigidity signal whose strength can be varied by partial restoration.

6 From Hard Classifier to Gradient Model

The first `IEG.CFG` heuristic used a hard classifier:

- high zero-word density in the body;
- high frequency of `0x42480000` (48.0) words;
- resulting classification: `sparse-fixed-schedule` or not.

This was useful as a first pass and yielded a measurable threshold. In the current restoration matrix, routine-preserving behavior remained intact through 61 restored cells and collapsed at 62 under the old model.

The next step replaced the pure classifier with a gradient:

- a continuous `ieg_rigidity_gradient`;
- a derived `routine_rigidity_signal`;
- a runtime `routine_hold` that decays over time and disruption.

This change preserved the broad distinction between the specimen routine path and the non-routine paths, but created a more realistic transitional band in which partial restorations retain reduced routine influence instead of falling instantly to zero.

7 Dynamic Probe Results

The most informative current probe family is time-based rather than single-event. Three probe types are especially important:

- routine endurance under repeated idle and disruption pressure;
- social endurance after a high-salience interaction;
- low-salience endurance after weaker interaction.

Probe family	0400	0400 + 1200	0400 + IEG
Routine endurance	breaks, sleeps	breaks, sleeps	preserves both cycles, stays awake
High-salience social endurance	breaks, sleeps	two social redirections, then sleep	preserves routine, stays awake
Low-salience social endurance	breaks, sleeps	weakens sooner, then sleep	preserves routine, stays awake

Table 1: Current dynamic boundary among the three main persistent-state paths.

The current interpretation is:

1. 0400 is a shutdown-oriented pivot;
2. 1200 supports a socially-cued, salience-sensitive persistence family;
3. specimen-side `IEG.CFG` supports a routine-preserving, multi-cycle persistence family.

8 Live-Hardware Test Matrix

The research has now progressed far enough to justify a disciplined live bench matrix. Four primary stick trees have been prepared:

1. baseline;
2. 0400;
3. 0400 + 1200;
4. 0400 + IEG.

Three live trial families are proposed:

1. no-touch routine pressure;
2. high-salience social pressure;
3. low-salience social pressure.

This matrix is designed to test whether the simulator-side distinction survives on real ERS-7 hardware and whether cue salience and multi-cycle persistence are observable outside the host-side model.

9 Contributions

This work makes five current contributions:

1. a verified recovery and preservation workflow for a repaired Sony-formatted ERS-7 MIND 2 stick;
2. a reproducible behavior-forensics workflow over preserved persistent state;
3. a host-side simulator that exposes candidate behavioral levers from persistent files;
4. a structural and dynamic characterization of specimen-side `IEG.CFG`;
5. a live-hardware matrix that turns current simulator conclusions into falsifiable bench trials.

10 Limitations

Several limits are important:

- the simulator is heuristic and not a decoded Sony runtime;
- current results are anchored in a small number of preserved reference trees and one especially valuable repaired specimen;
- the strongest findings are still host-side until repeated live-hardware tests confirm or refute them;
- some dynamic distinctions may reflect model choices rather than robot truth and must therefore be treated as hypotheses.
- current evidence supports live WLAN services and bounded content channels, but not a general claim of arbitrary in-place retail stick reprogramming over WLAN.

11 Next Steps

The next research steps are straightforward:

1. execute the prepared four-stick live-hardware matrix;
2. log repeated observations with the current observation template;
3. compare live results to the current dynamic verdict tables;
4. refine the model only where live evidence demands it;
5. extend the paper with real-hardware results, figures, and stronger bibliographic grounding in Sony documentation and robot behavior literature.

12 Conclusion

The present work establishes that AIBO ERS-7 MIND 2 behavior can be approached as a forensics problem over preserved filesystem state. A repaired and preserved working stick can serve as a behavioral specimen. Host-side modeling can isolate candidate persistent-state levers. Most importantly, current evidence suggests that persistent behavior is not monolithic: shutdown resistance, socially-cued persistence, and routine-preserving persistence are separable enough to support structured experimentation. That is a meaningful step toward a publishable science of legacy robotic behavior preservation and reconstruction.